

Imprecise probability in statistical inference

why it's needed, where it comes from, and how it helps

Ryan Martin¹

North Carolina State University

`www4.stat.ncsu.edu/~rgmarti3`

Finnish Statistical Society

Statistical Days 2024

Aalto University, Finland

June 3rd, 2024

¹Research partially supported by the NSF, SES-2051225

How can a discipline, central to science and to critical thinking, have two methodologies, two logics, two approaches that frequently give substantially different answers to the same problems? —Fraser

Unlike most philosophical arguments, this one has important practical consequences. The two philosophies represent competing visions of how science progresses and how mathematical thinking assists in that progress. —Efron

- If statisticians can't resolve these issues, then who are we?
- With 100+ years of experience, we should be the authorities on uncertainty, but instead, e.g.,

2021 ASA President's Task Force statement: *Different measures of uncertainty can complement one another*

- Lack of answers → confusion and distrust
- Not having the answers is OK, but...

Is complacency in the face of contradiction acceptable for a central discipline of science? —Fraser

- Resolving these issues is important
- I have new ideas...
- If you think I'm wrong, then please tell me
- In case I'm right, I hope the “new generation” is listening!

A new scientific truth does not triumph by convincing its opponents... but rather because its opponents eventually die, and a new generation grows up that is familiar with it. —Planck

- Basic claims:
 - probability theory is lacking (in certain contexts)
 - reliable inference/UQ requires something different
 - with *imprecise probability*, reliability is *possible*
- FAQs related to these claims:
 - *why imprecision?*
 - *where does imprecision come from?*
 - *what kind of imprecision?*
 - *how to achieve this imprecision?*
 - *how is imprecision beneficial?*
- This talk aims to answer these questions
- Suggested question topics are highlighted in green

- A simple textbook example hits on key issues
- Cancer screening
 - two relevant events

$$\underbrace{T = \{\text{test positive}\}}_{\text{data}} \quad \text{and} \quad \underbrace{H = \{\text{has cancer}\}}_{\text{hypothesis}}$$

- model: $P(T | H) = 0.95$ and $P(T | H^c) = 0.05$
 - prior $P(H) = 0.01$
- *Does a patient who tests positive have cancer or not?*
- Polarizing example
 - non-Bayesians infer H
 - Bayesians infer H^c

- Non-Bayesian solution:

- *law of likelihood*

$$P(T | H) \div P(T | H^c) = 19 \gg 1 \implies \text{infer } H$$

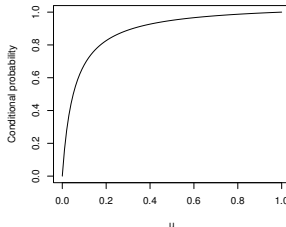
- p-value, $\pi_T(H) = 1 \implies \text{infer } H$

- Bayesian solution:

- prior to posterior update is

$$u \mapsto \frac{0.95 u}{0.95 u + 0.05 (1 - u)}$$

- if $u = 0.01$, then $P(H | T) = 0.16$
 - $0.16 \ll \frac{1}{2} \implies \text{infer } H^c$



Warm-up, cont.

- If we absolutely know $P(H)$, then use that info
- If we absolutely don't know $P(H)$, then
 - absurd to just take a wild guess
 - better to report the range of $P(H \mid T)$ as prior varies
 - this “Bayes” solution agrees with the p-value!
- Most likely case: we know *something* about $P(H)$
 - e.g., “we’re pretty sure $P(H) \in [a, b]$ ”
 - literature is silent about this *most likely case*!

Manski's Law of Decreasing Credibility.

Credibility of inference decreases with the strength of assumptions

- Data $Y \in \mathbb{Y}$, observed value y
- Model $\{P_\theta : \theta \in \mathbb{T}\}$
 - uncertain true value is Θ
 - generic values denoted by θ
- For now, assume no prior info about Θ is available

Goal.

Inference/uncertainty quantification about Θ , given $Y = y.....$

Statisticians want numerical measures of the degree to which data support hypotheses. —Hacking

- UQ about Θ , given $Y = y$?
 - first thought: “numerical measures of...” $\iff$ probability
 - i.e., return a “posterior” $(\Theta \mid Y = y) \sim Q_y$
- But: *probability doesn't provide reliable UQ in this context*

[The likelihood function] is not however a probability and does not obey the laws of probability; it involves no differential element $d\theta$...; it does none the less afford a rational basis for preferring some values of θ ... to others. —Fisher

- Fisher's warnings "likelihood $\neq$ probability" are familiar
- His point here:² for a hypothesis " $\Theta \in H$ "
 - "no differential element" $\implies$ size of H has no bearing
 - only *relative likelihood* values on H are relevant

²M., arXiv:2303.17425

Pushing likelihood too far, cont.

- Default prior Bayes (and others):
 - neither prior nor likelihood has a meaningful “ $d\theta$ ”
 - so how can posterior’s “ $d\theta$ ” be meaningful?

Illustrative example.

- Model: $Y \sim N(\Theta, 1)$
- Data: $y = 7$
- Default-prior Bayes posterior: $Q_y = N(7, 1)$
- Two hypotheses: $H_{\text{small}} = (6.5, 7] \subset (-\infty, 7] = H_{\text{large}}$
- Clearly, $Q_y(H_{\text{large}}) \gg Q_y(H_{\text{small}})$
- Does likelihood alone justify the conclusion that data supports H_{large} significantly more than H_{small} ?

Imprecision: *why?*

- Pushing likelihood too far has consequences
- In particular, the following would be problematic:
 - for some false hypotheses $H \not\in \Theta$
 - the rv $Y \mapsto Q_Y(H)$ tends to be large under P_Θ
- *False confidence*³ is a particular form of unreliability that all probabilistic UQ suffers from

False confidence theorem.

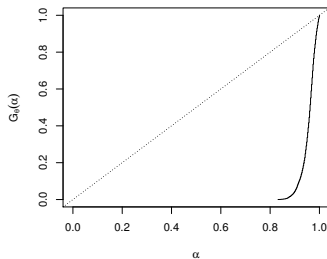
Let Q_Y be *any* data-dependent probability for Θ . For any $\alpha \in (0, 1)$, **there exists** hypotheses $H \subset \mathbb{T}$ such that

$$H \not\in \Theta \quad \text{and} \quad P_\Theta\{Q_Y(H) > 1 - \alpha\} > \alpha.$$

³Balch, M., & Ferson 2019, *Proc Roy Soc A*, arXiv:1706.08565

Imprecision: *why?*, cont.

- Example: Fieller–Creasy problem (simplest version)
 - model: $Y \sim N_2(\Theta, I)$, with $\Theta = (\Theta_1, \Theta_2)$
 - standard posterior: $Q_Y = N_2(y, I)$.
- Let $H = \{\theta : \theta_1/\theta_2 \leq 9\}$
- Suppose $\Theta = (1, 0.1)$, so that H is false
- We'd want/expect $Q_Y(H)$ to be small, but...



[Fisher] was the world's master of quantifying uncertainty, having developed many of the... procedures for doing so. —Pearl

- If not (precise) probability, then what?
- My answer: *imprecise probability*
- The need for imprecision wasn't lost on Fisher
 - *[A p-value]... does not justify any exact probability statement*
 - *no exact probability statements can be based on [conf. sets]*
- But he failed to formulate a suitable, general theory
- Fisher's "biggest blunder" = fiducial inference...

There is still more to be learned from Fisher, even when he seems clearly wrong, than from any other contributor to statistical thinking. —Dawid

- Not getting the “fiducial argument” right isn’t a big deal
- Incredibly ambitious goal
- Fisher’s efforts had huge impact
 - Dempster → origins of imprecise prob
 - Neyman → confidence sets, **inherently imprecise**
 - ...
- Not the end of the story...

Imprecision: *where?*

- If data + model can't reliably support precise prob's, then...
- Data-dependent *imprecise probabilities* for reliable UQ!

Definition — *inferential model* (IM).

- given data y , model, etc
- to every $H \subseteq \mathbb{T}$, assign a pair of numbers

$$\underline{\Pi}_y(H) \quad \text{and} \quad \overline{\Pi}_y(H)$$

- y -dependent measures of support for and plausibility of H

- *Possibility theory* is the simplest imprecise probability
- Similar to probability theory
 - determined by a “density function”
 - but with a different calculus — optimization vs integration
 - no differential element!
- Specifically:
 - possibility contour $\theta \mapsto \pi_y(\theta)$ with $\sup_{\theta} \pi_y(\theta) = 1$
 - plausibility/upper probability

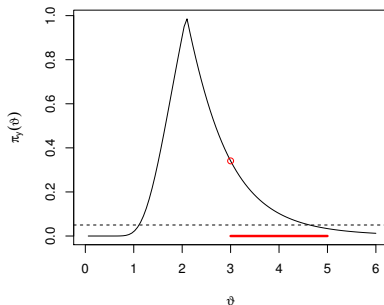
$$\overline{\Pi}_y(H) = \sup_{\theta \in H} \pi_y(\theta), \quad H \subseteq \mathbb{T}$$

- (complementary) support/lower probability

$$\underline{\Pi}_y(H) = 1 - \overline{\Pi}_y(H^c)$$

Imprecision: *what?*, cont.

- Illustration...
- Possibility contour plot
- Hypothesis $H = [3, 5]$
- $\bar{\Pi}_y(H)$ via optimization
 - no differential element
 - size of H is irrelevant



- Special case of a general construction
- Workhorse: *imprecise-probability-to-possibility transform*
- In the “no-prior” case, this takes a simple form:
 - let $R(y, \theta) = L_y(\theta) / \sup_{\vartheta} L_y(\vartheta)$ be relative likelihood
 - define a possibility contour

$$\pi_y(\theta) = P_{\theta}\{R(Y, \theta) \leq R(y, \theta)\}, \quad \theta \in \mathbb{T}$$

- then $\overline{\Pi}_y(H) = \sup_{\theta \in H} \pi_y(\theta)$, etc.
- Not unfamiliar... p-values!⁴
- Connection to/correction of Fisher's fiducial argument⁵

⁴M., arXiv:1203.6665 and arXiv:1511.06733

⁵M., arXiv:2303.08630

It is unacceptable if a procedure... of representing uncertain knowledge would, if used repeatedly, give systematically misleading conclusions. —Reid & Cox

Strong validity theorem.

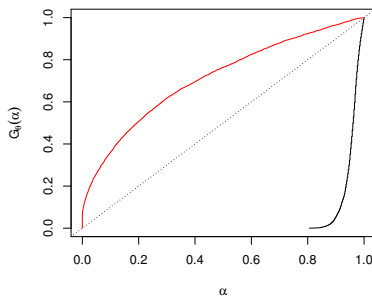
$$\sup_{\Theta} P_{\Theta} \{ \pi_Y(\Theta) \leq \alpha \} \leq \alpha, \quad \alpha \in [0, 1]$$

■ Remarks:

- most basic property, familiar for the no-prior case
- implies no false confidence

Imprecision: *how?*, cont.

- Same Fieller–Creasy simulation as before
- Compare CDFs of $Q_Y(H)$ and $\underline{\Pi}_Y(H)$
- H is false, so validity implies CDF is above diagonal
- False confidence vs. **no false confidence**



Corollary.

- 1 the test “reject H iff $\overline{\Pi}_Y(H) \leq \alpha$ ” satisfies

$$\sup_{\Theta \in H} P_{\Theta} \{ \overline{\Pi}_Y(H) \leq \alpha \} \leq \alpha, \quad \alpha \in [0, 1], \quad H \subseteq \mathbb{T}$$

- 2 the region $C_{\alpha}(y) = \{ \theta : \pi_y(\theta) > \alpha \}$ satisfies

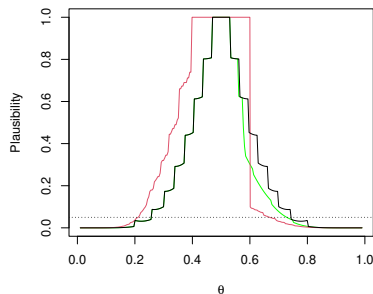
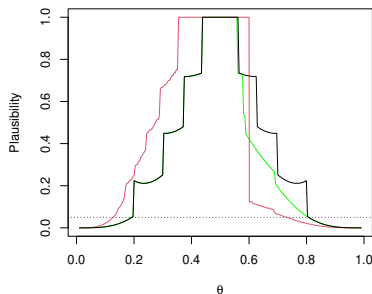
$$\sup_{\Theta} P_{\Theta} \{ C_{\alpha}(Y) \not\supseteq \Theta \} \leq \alpha, \quad \alpha \in [0, 1]$$

- These are the frequentist guarantees⁶ we know & love
- Property 1 is related to
 - Dawid's *production principle*
 - Walley's *fundamental frequentist principle*

⁶There's also a sort of “converse” (M., arXiv:2112.10904)

Example: one binomial

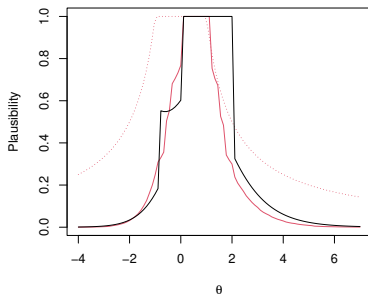
- $P_\theta = \text{Bin}(n, \theta)$
- Plots of the contour $\theta \mapsto \pi_y(\theta)$ — focus on black curve⁷
- Contours: $n = 8$ (left) and $n = 8$ (right), $\hat{\theta} = 0.5$



⁷Red & green: partial prior IMs that assume “90% sure that $\Theta \leq 0.6$ ”

Example: two binomials

- Clinical trial:
 - association between treatment & control?
 - let Θ denote the log odds ratio⁸
- Data $(n_1, y_1) = (43, 1)$ and $(n_2, y_2) = (39, 2)$
- Focus on the black curve only⁹



⁸Nuisance parameters eliminated via **conditioning** (M., arXiv:2309.13454)

⁹Red is a partial prior IM that assumes “ $E|\Theta| \leq 1$ ”

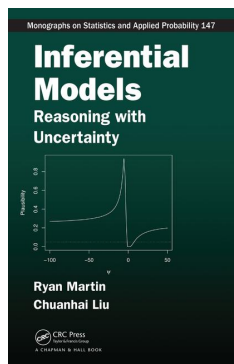
Example: Behrens–Fisher

- Difference of normal means, general variances¹⁰
- IM's validity is guaranteed, sim to check efficiency:
 - difficult unbalanced case, $(n_1, n_2) = (2, 20)$
 - compare coverage prob of 90% confidence intervals

Method	Coverage Prob
Hsu–Scheffe	0.9738
Jeffreys	0.9296
Ghosh & Kim	0.7873
Welch	0.8362
1st order	0.7399
Fraser et al	0.8617
IM	0.9082

¹⁰Nuisance par's eliminated via [marginalization](#) (M., arXiv:2309.13454)

- First book →
- New n -part series of working papers:
*Valid and efficient imprecise-probabilistic
inference with partial priors*
- $n = 3$ parts so far
 - 1 arXiv:2203.06703
 - 2 arXiv:2211.14567
 - 3 arXiv:2309.13454
- Working towards a new book focused on the above *possibilistic* formulation



■ New asymptotic results (arXiv:2404.15843)

**Large-sample theory for inferential models: a
possibilistic Bernstein–von Mises theorem**

Ryan Martin and Jonathan P. Williams

■ New computational strategy (arXiv:2404.19224)

**Variational approximations of possibilistic
inferential models**

Leonardo Cella and Ryan Martin

■ More on false confidence (arXiv:2404.16228)

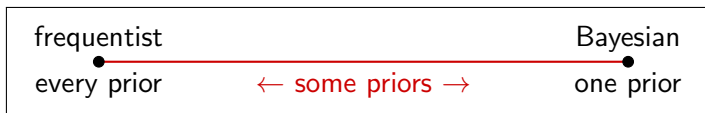
**Which statistical hypotheses are afflicted with
false confidence?**

- IMs & decision theory (arXiv:2112.13247)
- Reliability–coherence balance (arXiv:2312.14912)
- Conformal prediction, ML, etc¹¹



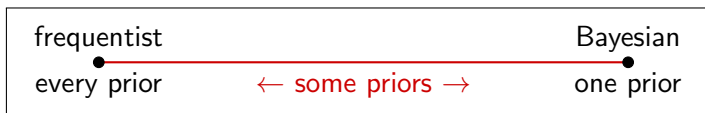
¹¹Work with my former PhD student, Leonardo Cella

Partial prior info and IMs



- This goes back to the warm-up example...
- There's a spectrum of problems indexed by prior info
- "Frequentist = every prior"¹²
- Key observations:
 - spectrum can't be described with ordinary probability
 - applying Bayes to a set of priors is inefficient (**dilation**)

¹²There's a **mathematical explanation** of this...



- Especially important in high-dim problems
- Main-stream methods for regularization are awkward:
 - Fs' penalties aren't part of the model/assumptions
 - Bs' priors impose more structure than is justified
- New idea:¹³
 - incorporates into the model exactly the structural assumptions that can be justified, à la Manski
 - provably reliable wrt assumed *imprecise prob model*

¹³M. arXiv:2211.14567

- Upper joint dist'n $\bar{P}_q$ for (Y, Θ) determined by
 - model/likelihood for $(Y \mid \Theta = \theta)$ as before
 - partial prior via possibility contour $q(\theta)$
- $\bar{\Pi}_y^q$ is a possibility measure, defined via optimization
 - contour is a **Choquet integral** with respect to $\bar{P}_q$

$$\begin{aligned}\pi_y^q(\theta) &= \bar{P}_q\{R_q(Y, \Theta) \leq R_q(y, \theta)\} \\ &= \int_0^1 \left[\sup_{\vartheta: q(\vartheta) > t} P_\vartheta\{R_q(Y, \vartheta) \leq R_q(y, \theta)\} \right] dt\end{aligned}$$

- $R_q(y, \theta) \propto L_y(\theta)q(\theta)$ is a *q-regularized* relative lik

Strong validity theorem.

$$\bar{P}_q\{\pi_Y^q(\Theta) \leq \alpha\} \leq \alpha, \quad \alpha \in [0, 1]$$

- Recall: $H = \{\text{cancer}\}$
- Practically realistic case: “ $P(H) \leq 0.05$ ” (say)
- Can encode this with a partial prior contour:

$$q(H) = 0.05 \quad \text{and} \quad q(H^c) = 1$$

Vacuous Prior

$$\underline{\Pi}_{\text{neg}}(H) = 0.00$$

$$\overline{\Pi}_{\text{neg}}(H) = 0.05 \implies \text{infer } H^c$$

$$\underline{\Pi}_{\text{pos}}(H) = 0.95 \implies \text{infer } H$$

$$\overline{\Pi}_{\text{pos}}(H) = 1.00$$

Partial Prior

$$\underline{\Pi}_{\text{neg}}^q(H) = 0.00$$

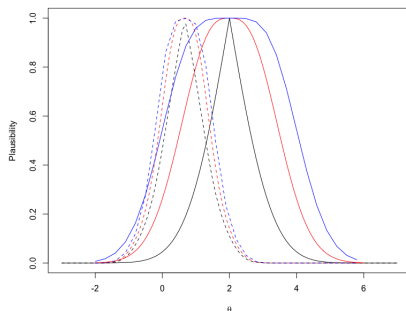
$$\overline{\Pi}_{\text{neg}}^q(H) < 0.01 \implies \text{infer } H^c$$

$$\underline{\Pi}_{\text{pos}}^q(H) = 0.00$$

$$\overline{\Pi}_{\text{pos}}^q(H) = 0.05 \implies \text{infer } H^c$$

Simple high-dimensional example

- Model: $Y_i \stackrel{\text{ind}}{\sim} N(\Theta_i, 1)$, $i = 1, \dots, n$
- *Sparsity* $\iff$ most θ_i 's are zero/small
- Prior possibility contour $q(\theta) = \exp(-c\|\theta\|^2)$
- Marginal contour for Θ_1 , with $y_1 = 2$ and $c = 1$
 - solid = vacuous prior, dashed = partial prior
 - $n = 1$, $n = 3$, and $n = 5$



Perhaps the most important unresolved problem in statistical inference is the use of Bayes theorem in the absence of prior information. —Efron

- Fisher had the right ideas but...
- Imprecision is imperative for reliable UQ
- New possibilistic IM framework:
 - likelihood-based
 - relatively simple
 - achieves strong validity
 - incorporates partial prior info

- Open methodological questions:
 - efficient Choquet integral computation?
 - partial prior elicitation?
 - high-dim problems?
 - possibilistic model assessment/selection?
- Open theoretical questions:
 - contraction–dilation comparisons?
 - decision-theoretic considerations?
 - revisiting classical **impossibility theorems** & paradoxes?
- Open philosophical questions:
 - **what makes the IM output meaningful is its properties?**
 - frequentist–Bayesian spectrum?
- Open applications: *all of them!*



ST 790 (001) Fall 2022 Advanced Special Topics

Imprecise-Probabilistic Foundations of Statistics & Data Science

[https://wordpress-courses2223.wolfware.ncsu.edu/
st-790-001-fall-2022/](https://wordpress-courses2223.wolfware.ncsu.edu/st-790-001-fall-2022/)

- Papers, talks, etc — www4.stat.ncsu.edu/~rmartin/
- Questions? — rgmarti3@ncsu.edu

Thanks for your attention!